

Ferrite Core Selection Using K_g Method

Input Parameters

- Inductance (L): $20\ \mu\text{H}$
- DC Current (I_o): $10\ \text{A}$
- AC Current (ΔI): $3\ \text{A}$
- Output Power (P_o): $120\ \text{W}$
- Regulation (α): 1%
- Switching Frequency (F_s): $100\ \text{kHz}$
- Operating Flux Density (B_m): $0.25\ \text{T}$
- Window Utilization (K_u): 0.4
- Temperature Rise Goal (T_r): 25°C

Calculations

1. Peak Current (I_{pk})

$$I_{pk} = I_o + \frac{\Delta I}{2} = 10 + \frac{3}{2} = \boxed{11.5\ \text{A}} \quad [[4]]$$

2. Energy-Handling Capability (E)

$$E = \frac{1}{2}LI_{pk}^2 = \frac{1}{2} \cdot 20 \times 10^{-6} \cdot (11.5)^2 \approx \boxed{0.00132\ \text{W-s}} \quad [[3]]$$

3. Electrical Condition Coefficient (K_e)

$$K_e = 0.145 \cdot P_o \cdot B_m^2 \cdot 10^{-4} = 0.145 \cdot 120 \cdot (0.25)^2 \cdot 10^{-4} \approx \boxed{2.175 \times 10^{-5}} \quad [[4]]$$

4. Core Geometry Coefficient (K_g)

$$K_g = \frac{E^2}{K_e \cdot \alpha} = \frac{(0.00132)^2}{2.175 \times 10^{-5} \cdot 1} \approx \boxed{0.0805\ \text{cm}^5} \quad [[5]][[6]]$$

Core Selection Criteria

Select a ferrite core with:

$$K_g \geq 0.0805\ \text{cm}^5 \quad (\text{from manufacturer datasheets}) \quad [[7]]$$

Summary of Results

Parameter	Value
Peak Current (I_{pk})	11.5 A
Energy-Handling (E)	0.00132 W-s
Electrical Coefficient (K_e)	2.175×10^{-5}
Core Geometry (K_g)	0.0805 cm ⁵